

Speaker: Sandipan Roy (University of Bath)

Date: 16/10/2026 at 13:15 in 4 West 1.7 (Wolfson Lecture Theatre)

Title: SASE: Squared Adjacency Spectral Embedding for Sparse Network Clustering

Abstract:

Spectral Clustering is a popular tool for identifying latent community structures in networks. However, many studies have shown that classical spectral clustering does not perform well on certain network structures, particularly core-periphery networks. To improve clustering performance in core-periphery structures, Adjacency Spectral Embedding (ASE) has been introduced, which performs clustering via a network's adjacency matrix instead of the graph Laplacian. Despite its advantages in this setting, the optimal performance of ASE is limited to dense networks, whilst network data observed in practice is often sparse in nature. To address this limitation, we propose a new approach which we term Squared Adjacency Spectral Embedding (SASE), motivated by the observation that the squared adjacency matrix will leverage the fewer connections in sparse structures more efficiently in clustering applications. Furthermore, we consider incorporating additional information in the form of node covariates, together with the network structure. We consider incorporating covariates before the spectral embedding process by combining covariates with the network representation. We also establish theoretical consistency through a bound on the misclustering rate under the stochastic block model, and demonstrate its clustering performance through simulation studies and real-data applications.